

Mitochondrion (pl. mitochondria)

A cell **organelle** involved in the production of energy for cell functions. It contains genetic material (mitochondrial DNA) derived solely from the mother.

Mitral valve

The valve that lies between the left **atrium** and left **ventricle** of the heart.

Motor neuron

A **nerve** cell that carries the impulses to muscles that cause movement.

Mucous membrane

The soft, mucus-secreting epithelial layer lining the tubes and cavities of the body.

Myocardium

The special muscle of the heart. The fibres form a network that can contract spontaneously.

Myofibril

Cylindrical element within muscle cells (fibres) consisting of thinner filaments that move to produce muscle contraction.

N**Nephron**

The **kidney's** filtering unit, consisting of a filtration capsule (glomerulus) and a series of tubules, that reabsorbs or excretes water and wastes to control fluid balance.

Nerve

Bundle of thread-like projections from individual **neurons** (nerve cells), held together by a fibrous sheath. Nerves carry electrical impulses to and from the brain and spinal cord and other body parts.

Neuron

A single **nerve** cell with long projections, the function of which is to transmit electrical impulses.

Nociceptor

A **nerve** ending responding to painful stimuli.

Nucleic acid

Deoxyribonucleic acid (**DNA**) or ribonucleic acid (**RNA**): chains of **nucleotides**, with genetic

information in the order of the bases of the **nucleotides**.

Nucleotide

Building-block subunit of a **nucleic acid** (**DNA**, **RNA**), consisting of a sugar, phosphate, and a nitrogen-containing **base**.

Nucleus (pl. nuclei)

Control centre of a cell, containing the genetic material **DNA**.

O**Oesophagus**

The muscular tube, also known as the gullet, that connects the **pharynx** with the stomach.

Oestrogen

A **sex hormone** that prepares the uterine lining for an implanted, fertilized egg and stimulates the development of a female's secondary sexual characteristics.

Olfactory nerve

One of two **nerves** of smell that run from the olfactory bulb in the roof of the nose directly into the underside of the brain.

Optic nerve

One of the two **nerves** of vision. Each one has about one million nerve fibres running from the **retina** to the brain, carrying visual stimuli.

Organ

Discrete body part or structure with a vital function: for example, the heart, **liver**, brain, or **spleen**.

Organelle

A tiny structure inside a cell that has a specific role. The **nucleus**, **mitochondrion**, and ribosomes are examples.

Ossicle

One of three tiny bones (the incus, malleus, and stapes) of the **middle ear** that convey vibrations from the **eardrum** to the inner ear.

Ossification

The process of formation, renewal, and repair of bone. Most bones in the body develop from **cartilage**.

Osteon

The rod-shaped unit, also called a Haversian system, that is the building block of cortical bone.

Ovary

One of two structures lying at the end of the **fallopian tubes** on each side of the **uterus**. They store ovarian follicles, release the mature **ova**, and produce the female **sex hormones** (**oestrogen** and **progesterone**).

Ovulation

The release of an **ovum** from a mature follicle in the **ovary** about midway through the menstrual cycle; if not fertilized, the egg is shed during menstruation.

Ovum (pl. ova)

The egg cell; if **fertilization** occurs, the ovum may implant in the **uterus** and develop into an **embryo**.

P**Pancreas**

A gland behind the stomach that secretes digestive **enzymes** and also **hormones** that regulate blood glucose levels.

Parasympathetic nervous system

One of the two divisions of the **autonomic nervous system**. It maintains and restores energy – for example, by slowing the rate and strength of the heartbeat or causing the liver to store glucose.

Parathyroid glands

Two pairs of yellowish **endocrine glands**, located behind the thyroid gland, that help to control the level of calcium in the blood.

Parotid glands

The large pair of salivary glands situated, one on each side, above the angles of the jaw just below and in front of the ears.

Pelvis

The basin-like ring of bones to which the lower end of the **spine** is attached and with which the thigh bones articulate. The term also refers to the general lower abdominal area.

Pericardium

The layers of membrane surrounding the heart. The outer fibrous sac encloses the